

Nirma University
Institute of Technology
Semester End Examination (IR), December-2025
M.C.A, Semester- I
6CS101CC25 Object Oriented Programming

Roll/
Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.
5. Assume suitable data wherever necessary and mention the same.

SECTION-I

Q-1 Write the output/errors of the following code snippets and justify your answer: Assume all necessary files are included. [18]

(A)
CO2
BL3

(1)

(6)

```
public static void main(String[] args) {  
    System.out.println("For Loop:");  
    for (int i = 55; i <= 100; i=i+11) {  
        System.out.println("i = " + i);  
    }  
}
```

(2)

```
public static void main(String[] args) {  
  
    for (int i = 1; i <= 10; i++) {  
        if (i == 7) continue;  
        System.out.println(i);  
    }  
}
```

(3)

```
public static void main(String[] args) {  
    int x = 200,y = 50, z = 100;  
  
    if(x > y && y > z){  
        System.out.println("Hello"); }  
    if(z > y && z < x){  
        System.out.println("Java"); }  
    if((y+200) > x && (y+150) < z){  
        System.out.println("Hello Java"); }  
}
```

- (B) Explain the importance of the Java Virtual Machine (JVM) in the Java programming. Why is the JVM essential for achieving Java's "write once, run anywhere" capability? (6)
CO1
BL2
- (C) Differentiate between the replace(), length() and substring() methods of the **String** class. (6)
CO2
BL4
- Q-2** **Answers the following:** [16]
(A) Explain the role of static method in Java programming. Discuss how static method differ from non-static method in terms of accessibility and usage. Support your explanation by providing a suitable program that demonstrates the declaration and invocation of a static method. (8)
CO2
BL5
- (B) Explain the concept of the else-if ladder in java with suitable example. How does it differ from a simple if-else statement? (8)
CO2
BL2
Write a java program using an else-if statement where the real-life scenario is used.
- Q-3** **Answers the following:** [16]
(A) Create a Java program to implement an **employee** class with fields like employee ID, name, and salary. Include methods to display employee details, and calculate salary using formula Gross Salary = Salary + HRA + DA (Assume necessary information if required) (8)
CO1
BL6
- (B) Explain how a constructor in Java differs from a method. Discuss the key differences in terms of name, return type, purpose, and invocation. Support your explanation with a suitable program that demonstrates the use of both a constructor and a method in a java class. (8)
CO2
BL6

SECTION-II

- Q-4** **Answers the following:** [18]
(A) What is JDBC (Java Database Connectivity) and what are JDBC Drivers? Explain how JDBC helps in developing client-server application in java programming. (6)
CO3
BL2
- (B) Differentiate between built-in packages and user-defined package in java. Also write an example of user-defined type package with real time application. (6)
CO2
BL4
- (C) Explain the process of creating a thread in Java using the **Thread** class. Describe the steps required to extend the Thread class and implement the run() method. (6)
CO3
BL5
- Q-5** **Answers the following:** [16]
(A) What is the significance of the abstract class and interface in Java programming? Also write a program that uses an interface named Bank containing a method to calculate (8)
CO2
BL5

interest. Different bank classes like SBI, ICICI, and HDFC implement this interface and provide their own interest rates. (Assume necessary information if required)

- (B) Is it compulsory to use exception handling in Java (8)
CO3 programming? Explain the try and catch block with a suitable
BL5 example which uses an exceptions ArithmeticException and
NullPointerException in java program.

- Q-6** **Answers the following:** [16]
(A) Differentiate between method overloading and method (8)
CO2 overriding in Java programming. Also write a Java program to
BL4 create a class Vehicle with a method called speedUp(). Create
two subclasses Car and Bicycle. **Override** the speedUp()
method in each subclass to increase the vehicle's speed
differently.

- (B) Identify the use of stream in context of Java I/O. Write a (8)
CO3 program in java to copy one binary file to other using File Input
BL3 Stream and Output Stream classes
